

Dennis Johan Loevlie

Multimodal foundation models for tabular data · probabilistic deep learning

+1 724 841 8769 · dennis.loevlie@tufts.edu · dennisloevlie.com · [Google Scholar](https://scholar.google.com/citations?user=...) · github.com/loevlie

EDUCATION

Centrum Wiskunde & Informatica & University of Amsterdam

Incoming, Aug 2026 –

PhD in Computer Science. Advised by **Madelon Hulsebos** (TRLab, CWI) and **Jan-Willem van de Meent** (AMLab, UvA); **ELLIS PhD Program** international co-advisor TBD. Focus: multimodal tabular foundation models; probabilistic deep learning.

Tufts University, Medford, MA

Sep 2024 – May 2026

M.S. in Computer Science. GPA: 3.96 / 4.00. [Transcript](#).

Carnegie Mellon University, Pittsburgh, PA

Sep 2019 – Dec 2020

M.S. in Chemical Engineering. GPA: 3.91 / 4.00.

West Virginia University, Morgantown, WV

Sep 2016 – Aug 2019

B.S. in Chemical Engineering, *cum laude*, with Honors.

PUBLICATIONS

Machine Learning

- E. Harvey*, **D. J. Loevlie***, A. A. Satani, W. Chen, D. M. Kent, and M. C. Hughes. “A Multi-Dataset Benchmark of Multiple Instance Learning for 3D Neuroimage Classification”. In: *Conference on Health, Inference, and Learning (CHIL) 2026*. *Equal contribution. [\[paper\]](#) [\[code\]](#). 2026. arXiv: [2604.26807](#).
- E. Harvey, **D. J. Loevlie**, and M. C. Hughes. “Synthetic Data Reveals Generalization Gaps in Correlated Multiple Instance Learning”. In: *ML4H 2025 Symposium (Findings Track)*. [\[paper\]](#). 2025. arXiv: [2510.25759](#).

Chemistry & Computational Science

- D. J. Loevlie**, B. Ferreira, and G. Mpourmpakis. “Demystifying the Chemical Ordering of Multimetallic Nanoparticles”. In: *Accounts of Chemical Research* 56.3 (2023). [\[code\]](#), pp. 248–257.
- M. Salem, **D. J. Loevlie**, and G. Mpourmpakis. “Single Atom Alloys Segregation in the Presence of Ligands”. In: *Journal of Physical Chemistry C* 127.46 (2023), pp. 22790–22798.
- R. Ding, I. M. Padilla Espinosa, **D. Loevlie**, S. Azadehramjbar, A. J. Baker, G. Mpourmpakis, A. Martini, and T. D. B. Jacobs. “Size-Dependent Shape Distributions of Platinum Nanoparticles”. In: *Nanoscale Advances* 4 (2022), pp. 3978–3986.
- A. V. Nagarajan, **D. J. Loevlie**, M. J. Cowan, and G. Mpourmpakis. “Resolving Electrocatalytic Imprecision in Atomically Precise Metal Nanoclusters”. In: *Current Opinion in Chemical Engineering* 36 (2022), p. 100784.

RESEARCH EXPERIENCE

Tufts University — Hughes Lab

Aug 2024 – Present

Graduate Researcher with Prof. Michael Hughes

- Co-led (equal contribution) the CHIL 2026 multi-dataset benchmark of multiple instance learning (MIL) for 3D neuroimage classification across 7 MRI/CT datasets and 9 tasks; showed a simple mean-pooling MIL baseline matches or outperforms recent attention-based MIL and 3D CNN methods on 4 of 6 moderate-sized tasks while being **25× faster to train**.
- Contributed to the ML4H 2025 study on a *generalization gap* between correlated and non-correlated MIL architectures: implemented SmAP (the newest NeurIPS MIL method) into the lab codebase, reduced a training-example bottleneck from *quadratic* to *linear scaling* to unlock experiments on a substantially larger semi-synthetic dataset, and helped design the synthetic benchmark with a Bayes-optimal upper bound.
- Designed convolutional and transformer-based classification pipelines on 3D MRI/CT for early indicators of dementia and stroke; trained on Tufts HPC with multi-GPU distributed setups.
- Developing a regularization method to encourage interpretable attention scores, and methods for supervised learning under noisy labels.

Tufts University — Sinapov Lab

Jan 2025 – Present

Graduate Researcher with Prof. Jivko Sinapov

- Trained Qwen2.5-Coder-7B with Group Relative Policy Optimization (GRPO) and a custom 3-part reward (structure, aesthetics, semantic alignment) to generate SVGs from text; achieved an **18% improvement** on a benchmark covering aesthetics, alignment, and code validity.
- Investigating why vision-language models (Qwen3-VL) and open-vocabulary segmentation models (SAM 3) fail at object counting in dense scenes; designing tool-use and trajectory-based methods drawing on cognitive-science accounts of human counting.

University of Pittsburgh — CANELa Lab

Jun 2021 – Jan 2023

Graduate Researcher with Prof. Giannis Mpourmpakis

- Proposed a weight-initialization method that delivered a **71% RMSE reduction** for predicting nanoparticle stability across compositions and atomic orderings (Loevlie *et al.*, *Acc. Chem. Res.* 2023).
- Ensured fair evaluation of ML architectures for material-property prediction; co-author on Salem *et al.*, *JPCC* 2023.
- Wrote the ML methods and background section for the Nagarajan *et al.* review (Curr. Opin. Chem. Eng., 2022).

Carnegie Mellon University — Kitchin Group

Dec 2019 – Dec 2020

Graduate Researcher with Prof. John Kitchin

- Trained a CNN classifier to extract experimental information from microscopy images; rebuilt Mathematica analysis tooling as fast, interactive Python.
- Released [nb_search](#), an open-source Python package for searching and indexing across Jupyter notebooks.

AWARDS & HONORS

- 2026 **1st place** (10 teams), *BrainStorm Neural Decoder Challenge* — real-time auditory decoding from 1024-channel ECoG; 95% accuracy, sub-ms inference, edge-deployable model.
- 2024 **Community Grant**, *Hugging Face* — for [Depth Anything](#) demonstration work on video.
- 2019 **1st place** (Advanced category), *AVEVA National Simulation Competition*.
- 2018 **2nd place**, *AIChE National Poster Competition*, Computing & Process Control Division.

INDUSTRY EXPERIENCE

Massachusetts General Hospital & Harvard Medical School

2025 – Present

Machine Learning Researcher — computational pathology

- **Systems:** building segmentation models for colon-cancer pathology slides at gigapixel resolution; pipeline integrates with clinical workflows for downstream evaluation.
- **Research:** dataset-construction methodology and collaboration with pathologists to ground model outputs in expert-validated labels.

KEF Robotics, Pittsburgh, PA

Jan 2024 – Aug 2024

Senior Computer Vision & ML Engineer

- **Leadership:** sole ML engineer leading a team of five on a one-year, \$500K project; owned task breakdown, budgeting, and applied-research direction.
- **Systems:** shipped on-device object detection with 3D localization, monocular depth estimation, and 3D map generation from monocular drone imagery; deployed on flight hardware via TensorRT. Two in-person demos secured the next round of funding.

KEF Robotics, Pittsburgh, PA

Jan 2023 – Dec 2023

Computer Vision Engineer

- **Research:** designed a three-stage multimodal semantic-segmentation pipeline for tethered-drone power-line detection under severe data constraints — (1) Mask2Former fine-tuned from COCO on drone simulation data, (2) progressive sim-to-real domain adaptation using limited labeled real-world RGB imagery, (3) automated cross-modal label transfer via geometric correspondence to synthesize IR training data from RGB predictions, removing the need for manual IR annotation.
- **Systems:** optimized the segmentation architecture for a **45% inference-speed gain** at < 1% accuracy cost; work featured in invited talk at *XChangeIdeas Pittsburgh* (1 of 2 KEF presenters; 100+ industry attendees).

Multimodal AI Intern (insurance-tech startup, MIT-led)

- **Systems:** integrated and tested learning-based duplicate detection and open-vocabulary instance segmentation in React / React Native.
- **Research:** prototyped retrieval and matching components on multimodal product data.

AiThElite

Dec 2020 – May 2021

Lead Data Scientist (collegiate-athletics AI startup)

- Built end-to-end data pipeline (Selenium scrapers → NumPy/Pandas feature engineering → scikit-learn models) and shipped a Django/AWS-hosted prediction product.

OPEN SOURCE

- [huggingface/transformers](#) – contributor; merged fix restoring torch.compile compatibility for Mask2Former and OneFormer
- [Franblueeee/torchmil](#) – contributor; added the ML4H 2025 synthetic dataset and Bayes-optimal upper bound
- [loevlie/neuropt](#) – author; LLM-guided neural architecture search and hyperparameter tuning
- [loevlie/gpt4readability](#) – author; README generation and code-improvement suggestions for any code repository, powered by OpenAI or a local Llama model
- [JohanDL/Depth-Anything-Video](#) – author; Hugging Face Community Grant recipient

SELECTED TALKS

Synthetic Data Reveals Generalization Gaps in Correlated Multiple Instance Learning. Poster, ML4H 2025 Symposium, San Diego, CA. Dec 2025.

Computer Vision for Unmanned Aerial Vehicles. Invited talk, XChangeIdeas Pittsburgh. 2023.

SERVICE & OUTREACH

Industry volunteer, youth robotics team building accessibility tools for blind soccer players 2024

TECHNICAL SKILLS

Languages Python · C++ · Java · JavaScript · MATLAB

ML / AI PyTorch · JAX · Hugging Face (Transformers, Datasets, TRL, PEFT) · GRPO / DPO · multimodal learning · RL post-training · multiple instance learning · computer vision

Engineering FSDP · DeepSpeed · vLLM · FlashAttention · Triton · CUDA · TensorRT · SLURM / HPC · multi-GPU distributed training · Docker · AWS